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**EXNO 2: Analysis of SWR (Standing Wave Ratio) Transmission Lines by
varying a) load impedance b) characteristic impedance**

Test Case 1:

Analyze the SWR of a transmission line for a fixed characteristic impedance (Z_0) of 50 ohms, as the load impedance (Z_L) is varied across 650 ohms to 850 ohms. Verify how the SWR changes with each value of Z_L . When $Z_L = Z_0$, the SWR should equal 1 (indicating no reflection).

AIM: To analyze the SWR of a transmission line for a fixed characteristic impedance (Z_0) of 50 ohms, as the load impedance (Z_L) is varied across the following values: 25 ohms, 50 ohms, 75 ohms, and 100 ohms. To verify how the SWR changes with each value of Z_L and when $Z_L = Z_0$.

SOFTWARE REQUIRED: SCILAB Version 6.1.1

PROCEDURE:

- Open SCILAB software
- Open file-new
- Put the input values
- Type the program
- Execute the program
- End the program

PROGRAM:

```
clc;
```

```
clear;
```

```

close;

Zo=200;

ZR=650-(%i*475);

K=(ZR-Zo)/(ZR+Zo);

ampK=sqrt((real(K)^2)+(imag(K)^2));S=(1+ampK)/(1-ampK);

printf("(c)Standing wave ratio = %f",round(S*1000)/1000);

```

OUTPUT:

Zo=50 ohms Variable: ZR

ZR Ohms	SWR
650-(%i*475)	19.969000
700-(%i*475)	20.469000
750-(%i*475)	21.036000
800-(%i*475)	21.657000
850-(%i*475)	22.323000

ZR=50/_40ohms Variable: Zo

Zo Ohms	SWR
200	4.170000
300	6.110000
400	8.081000
500	10.065000
600	12.054000

RESULT:

The SWR of a transmission line for a fixed characteristic impedance (Zo) of 50 ohms, is analyzed for various values of load impedance ZL. SWR is also examined for the condition ZL=Zo.

Test Case 2:

Analyze the SWR for a fixed load impedance (Z_L) of 50 ohms while varying the characteristic impedance (Z_0) of the transmission line to 250 ohms to 400 ohms. Verify how the SWR changes with each value Z_0 including $Z_0=Z_L$.

AIM: To analyze the SWR for a fixed load impedance (Z_L) of 75 ohms while varying the characteristic impedance (Z_0) of the transmission line to 25 ohms, 50 ohms, 75 ohms, and 100 ohms. To Verify how the SWR changes with each value Z_0 including $Z_0=Z_L$

SOFTWARE REQUIRED:

SCILAB Version 6.1.1

PROCEDURE:

- Open SCILAB software
- Open file-new
- Put the input values
- Type the program
- Execute the program
- End the program
- End the program

PROGRAM:

```
clc;
```

```
clear;
```

```
close;
```

```
Zo=200;ZR=650-(%i*475);
```

```

K=(ZR-Zo)/(ZR+Zo);

ampK=sqrt((real(K)^2)+(imag(K)^2));S=(1+ampK)/(1-ampK);

printf("(c)Standing wave ratio = %f",round(S*1000)/1000);

```

OUTPUT:

ZR=50/_40o ohms Variable: Zo

Zo Ohms	SWR
200	4.170000
300	6.110000
400	8.081000
500	10.065000
600	12.054000

RESULT:

SWR for a fixed load impedance (ZL) of 50 ohms is analyzed while varying the characteristic impedance (Z0) of the transmission line to 250 ohms to 400 ohms. Impact of the condition Z0=ZL on SWR is also examined for various values of Z0 .